

Ring of Algebraic Integers

Given a field K containing the field $\mathbb{Q}$ of rational numbers, recall that an element α in K is called an algebraic integer if there is a *monic* polynomial f in $\mathbb{Z}[X]$ such that $f(\alpha) = 0$.

Matrix representation

Given that the field K is a finite dimensional vector space over $\mathbb{Q}$, let $S = \{v_1, \dots, v_n\}$ be a basis of K over $\mathbb{Q}$. For any element α of K , we represent multiplication by α on K in terms of the basis S as an $n \times n$ matrix $M_S(\alpha)$ with entries in $\mathbb{Q}$:

$$\alpha \cdot v_i = \sum_{j=1}^n M_S(\alpha)_{i,j} v_j$$

One checks that $\alpha \mapsto M_S(\alpha)$ gives us a ring homomorphism from K to the ring $M_n(\mathbb{Q})$ of $n \times n$ matrices with entries in $\mathbb{Q}$. Moreover, it is clear that this homomorphism is injective.

For each i , the matrix $M_S(v_i)$ has entries in $\mathbb{Q}$. Since S is finite, there is a non-zero integer k such that, for each i , the matrix $M_S(kv_i)$ has entries in $\mathbb{Z}$.

Further note that if $T = \{kv_1, \dots, kv_n\}$, then $M_T(\alpha) = M_S(\alpha)$ for all α in K .

There is a basis $T = \{w_1, \dots, w_n\}$ of K over $\mathbb{Q}$ such that, for each i , the matrix $M_T(w_i)$ has entries in $\mathbb{Z}$.

Order

For a basis T of the vector space K over $\mathbb{Q}$, we define R_T to be the subset of K that consists of those α for which $M_T(\alpha)$ has entries in $\mathbb{Z}$. Since $M_n(\mathbb{Z})$ is a subring of $M_n(\mathbb{Q})$ it follows that R_T is a subring of K . Moreover, since R_T is a subgroup of the free group $M_n(\mathbb{Z})$, it is also a free group. In other words, there is a set $U = \{u_1, \dots, u_m\}$ of elements of R_T such that $R_T = \oplus_{k=1}^m \mathbb{Z}u_k$.

As seen above, we may assume that R_T contains T . Hence, every element of T is a *unique* $\mathbb{Z}$ -linear combination of the elements of U . It follows that $m = n$ and U is *itself* a basis of K .

If $M_U(\alpha)$ is an integer matrix then $\alpha \cdot R_T \subset R_T$ so that $\alpha = \alpha \cdot 1$ lies in R_T . It follows easily that $R_T = R_U$.

In summary, we have shown that if T is a basis of K over $\mathbb{Q}$, the ring R_T of elements α in K such that $M_T(\alpha)$ is an integer matrix is of the form $R_T = R_U$, where U is a basis of K over $\mathbb{Q}$.

A subring R of K which is the $\mathbb{Z}$ -linear span of a basis of K over $\mathbb{Q}$ is called an *order* in K .

Trace and Norm

Given bases $S = \{v_1, \dots, v_n\}$ and $T = \{w_1, \dots, w_n\}$ of K over $\mathbb{Q}$, let G be the invertible matrix such that $w_i = \sum_{j=1}^n G_{i,j} v_j$. It follows that

$$\begin{aligned} \alpha \cdot w_i &= \sum_{j=1}^n G_{i,j} \alpha \cdot v_j &= \sum_{j=1}^n \sum_{k=1}^n G_{i,j} M_S(\alpha)_{j,k} v_k \\ \alpha \cdot w_i &= \sum_{j=1}^n M_T(\alpha)_{i,j} w_j &= \sum_{j=1}^n \sum_{k=1}^n M_T(\alpha)_{i,j} G_{j,k} v_k \end{aligned}$$

Hence $G \cdot M_S(\alpha) = M_T(\alpha) \cdot G$.

It follows that $\text{Trace}(M_S(\alpha))$ and $\det(M_S(\alpha))$ are independent of the choice of basis. So, we define

$$\begin{aligned} \text{Tr}_{K/\mathbb{Q}}(\alpha) &= \text{Trace}(M_S(\alpha)) \\ \text{Nm}_{K/\mathbb{Q}}(\alpha) &= \det(M_S(\alpha)) \end{aligned}$$

Similarly, the characteristic polynomial $\text{ch}_{K/\mathbb{Q}}(\alpha)(X) = \det(X - M_S(\alpha))$ is independent of the choice of basis; it is a monic polynomial with coefficients in the subring of $\mathbb{Q}$ generated by the entries of $M_S(\alpha)$.

By the Cayley-Hamilton theorem, we see that α satisfies the polynomial $\text{ch}_{K/\mathbb{Q}}(\alpha)$.

Note that if there is *some* basis S of K in which $M_S(\alpha)$ has integer coefficients, then

- $\text{Tr}_{K/\mathbb{Q}}(\alpha)$ and $\text{Nm}_{K/\mathbb{Q}}(\alpha)$ are integers,
- $\text{ch}_{K/\mathbb{Q}}(\alpha)$ has integer coefficients, and
- α is an algebraic integer.

In particular, we note that the elements of an order are algebraic integers.

Inner product

Given $\alpha \neq 0$, we see that $\text{Tr}_{K/\mathbb{Q}}(\alpha \cdot \alpha^{-1}) = n \neq 0$. It follows that the pairing

$$(\cdot, \cdot) : K \times K \rightarrow \mathbb{Q} \text{ given by } (\alpha, \beta) = \text{Tr}_{K/\mathbb{Q}}(\alpha \cdot \beta)$$

is a non-degenerate bilinear symmetric pairing on K as a vector space over $\mathbb{Q}$.

Thus, for $S = \{v_1, \dots, v_n\}$ a basis of K , the matrix B_S with entries $(B_S)_{i,j} = (v_i, v_j)$ is an invertible symmetric matrix over $\mathbb{Q}$.

The Dual of an order R

Given an order R in K , we define the *dual* as the collection R^* of all $\alpha \in K$ such that $\text{Tr}_{K/\mathbb{Q}}(\alpha R) \subset \mathbb{Z}$.

Note that R is contained in R^* .

Now, suppose that $S = \{v_1, \dots, v_n\}$ is a $\mathbb{Z}$ -basis of the order R . Since R is a ring, $v_i \cdot v_j$ are in R . Hence, they are algebraic integers. Thus the matrix B_S has integer entries.

If $\alpha \in R^*$ is expressed as $\alpha = \sum_{i=1}^n a_i v_i$, then we have

$$(\alpha, v_j) = \sum_{i=1}^n a_i (B_S)_{i,j}$$

We see that R^* can be identified with the collection of vectors $\mathbf{a} = (a_i)$ in $\mathbb{Q}^n$ such that $B_S \mathbf{a}$ is a vector with integer entries. In other words, it can be identified with $(B_S)^{-1} \mathbb{Z}^n$ with respect to the basis S .

In particular, we note that R^* is *also* the $\mathbb{Z}$ -linear span of a basis and is contained in $(1/\det(B_S))R$.

If $S = \{v_1, \dots, v_n\}$ is a basis of on order R then R^* contains R and is contained in the subgroup $(1/\det(B_S))R$.

This result will be useful once we prove that all algebraic integers are contained in R^* .

One algebraic integer

Given an algebraic integer α in K , let f be the monic polynomial in $\mathbb{Z}[X]$ that is satisfied by α . We get a natural homomorphism $\mathbb{Z}[X]/\langle f \rangle \rightarrow K$ which sends X to α . The image of this homomorphism is a domain which is the subring of K generated by α ; we denote it by $\mathbb{Z}[\alpha]$.

Since $\mathbb{Z}[\alpha]$ is a domain there is an irreducible factor p of f in $\mathbb{Z}[X]$ which is satisfied by α . By Gauss' lemma, we see that p is the minimal polynomial of α and that p is monic as well. This shows that $\mathbb{Z}[X]/\langle p \rangle$ is isomorphic to $\mathbb{Z}[\alpha]$ which is freely generated over $\mathbb{Z}$ by the elements $1, \alpha, \dots, \alpha^{d-1}$ where d is the degree of p .

Note that $1, \alpha, \dots, \alpha^{d-1}$ is a $\mathbb{Q}$ -basis for $\mathbb{Q}[\alpha]$ which is a subfield of K . Let $u_1, \dots, u_m$ be a basis of K over this subfield $\mathbb{Q}[\alpha]$.

We then check that $S = \{\alpha^i u_j : i = 0, \dots, d-1; j = 1, \dots, m\}$ form a basis for K over $\mathbb{Q}$.

With respect to this basis, we see that $M_S(\alpha)$ has a block diagonal form with m diagonal blocks each of which is the Cayley matrix of p :

$$\begin{pmatrix} 0 & 0 & \dots & 0 & -p_0 \\ 1 & 0 & \dots & 0 & -p_1 \\ 0 & 1 & \dots & 0 & -p_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -p_{d-1} \end{pmatrix}$$

where $p(X) = X^d + p_{d-1}X^{d-1} + \cdots + p_1X + p_0$ where p_i are integers. It follows that $M_S(\alpha)$ has integer entries, so α lies in R_S .

Every algebraic integer lies in some order R in K .

We now expand this calculation to a *pair* of algebraic integers.

Two algebraic integers

Given algebraic integers α and β in K that satisfy the monic polynomials f and g in $\mathbb{Z}[X]$, we consider the subring $\mathbb{Z}[\alpha, \beta]$ of K generated by them. This ring is the image of the homomorphism $\mathbb{Z}[X, Y] \rightarrow K$ that sends X to α and Y to β . In fact, it is the image of

$$\frac{\mathbb{Z}[X, Y]}{\langle f(X), g(Y) \rangle} \rightarrow K$$

since $f(X)$ and $g(Y)$ lie in the kernel of the above homomorphism.

Note that this homomorphism is *not*, in general, an inclusion, *even if* f and g are the minimal polynomials for α and β respectively. However, we note that the map extends to a map

$$\frac{\mathbb{Q}[X, Y]}{\langle f(X), g(Y) \rangle} \rightarrow K$$

and $\mathbb{Z}[X, Y]/\langle f(X), g(Y) \rangle$ is a subring of $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$.

A basis. Given a polynomial $t(X, Y)$ we can think of it as a polynomial in X with coefficients in $\mathbb{Z}[Y]$. Since $f(X)$ is monic, we can write

$$t(X, Y) = q_1(X, Y)f(X) + s(X, Y)$$

where $s(X, Y)$ when considered as a polynomial in X with coefficients in $\mathbb{Z}[Y]$ has degree *less than* the degree of f .

Now, we can think of the polynomial $s(X, Y)$ as a polynomial in Y with coefficients in $\mathbb{Z}[X]$. Since $g(Y)$ is monic, we can write

$$s(X, Y) = q_2(X, Y)g(Y) + r(X, Y)$$

where $r(X, Y)$ when considered as a polynomial in Y with coefficients in $\mathbb{Z}[X]$ has degree *less than* the degree of g . Moreover, the degree of these coefficients considered as polynomials in X is less than the degree of the $s(X, Y)$ as a polynomial in X .

It follows that $r(X, Y)$ is an integer linear combination of monomials of the form $X^i Y^j$ where i is less than the degree of f and j is less than the degree of g .

In other words, if the degree of f is d and the degree of g is e , we can write every polynomial $t(X, Y)$ as

$$t(X, Y) = q_1(X, Y)f(X) + q_2(X, Y)g(Y) + \sum_{i=0, j=0}^{i=d-1, j=e-1} a_{i,j} X^i Y^j$$

By comparison of degrees with respect to X and Y on the two sides, we see that the $a_{i,j}$ are uniquely determined elements of $\mathbb{Z}$. In other words, we can identify the *additive groups*

$$\frac{\mathbb{Z}[X, Y]}{\langle f(X), g(Y) \rangle} = \bigoplus_{i=0, j=0}^{i=d-1, j=e-1} \mathbb{Z} \cdot (X^i Y^j)$$

Note that $\{X^i Y^j\}_{i,j=0}^{i=d-1, j=e-1}$ form a basis of the vector space $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$.

The ring $\mathbb{Z}[X, Y]/\langle f(X), g(Y) \rangle$ is the $\mathbb{Z}$ -linear span of the basis $\{X^i Y^j\}_{i,j=0}^{i=d-1, j=e-1}$ of the $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$.

Multiplication operators. Suppose that $f(X) = X^d + \sum_{i=0}^{d-1} f_i X^i$. Then multiplication by X is a linear transformation of $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$ given by

$$\begin{aligned} X^i Y^j &\mapsto X^{i+1} Y^j \text{ for } i < d-1 \\ X^{d-1} Y^j &\mapsto -\sum_{i=0}^{d-1} f_i X^i Y^j \end{aligned}$$

In particular, in the above basis it is given by multiplication by a matrix M_X which has integer coefficients.

Similarly, multiplication by Y is given by multiplication by a matrix M_Y which has integer coefficients.

Thus, multiplication by $X + Y$ (respectively $X \cdot Y$) on $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$ is given by matrix multiplication by $M_X + M_Y$ (respectively $M_X \cdot M_Y$) which is a matrix with integer coefficients.

By the Cayley-Hamilton theorem, we see that the element $X + Y$ (respectively $X \cdot Y$) of the *ring* $\mathbb{Q}[X, Y]/\langle f(X), g(Y) \rangle$ satisfies the polynomial $p(T) = \det(T - (M_X + M_Y))$ (respectively $q(T) = \det(T - (M_X \cdot M_Y))$) which is a *monic* polynomial with integer coefficients.

Since $\alpha + \beta$ (respectively $\alpha \cdot \beta$) is the *image* of $X + Y$ (respectively $X \cdot Y$) in K under the above homomorphism, we see that it also satisfies $p(T)$ (respectively $q(T)$).

The sum $\alpha + \beta$ and product $\alpha \cdot \beta$ of algebraic integers α and β in K are *also* algebraic integers.

The ring of algebraic integers in K

We use $\mathcal{O}_K$ to denote the *subset* of K that consists of algebraic integers. By what we have proved above, we see that $\mathcal{O}_K$ is closed under addition and multiplication. It is clear that 1 and 0 are algebraic integers. If α is an algebraic integer, then it is clear that $-\alpha$ is an algebraic integer.

The collection $\mathcal{O}_K$ of algebraic integers in K forms a subring of K .

We have seen that, if R is an order in K , then it consists of algebraic integers, so $R \subset \mathcal{O}_K$.

We have also seen that if α is an algebraic integer then $\text{Tr}_{K/\mathbb{Q}}(\alpha)$ lies in $\mathbb{Z}$.

So, if α is an element of $\mathcal{O}_K$, then $\alpha \cdot R \subset \mathcal{O}_K$ implies that $\text{Tr}_{K/\mathbb{Q}}(\alpha \cdot R) \subset \mathbb{Z}$. In other words, $\alpha \in R^*$.

This means that $\mathcal{O}_K \subset R^*$; so $\mathcal{O}_K$ is “trapped” between the free abelian groups R and $R^* \subset (1/N)R$ for some positive integer N . Thus, $\mathcal{O}_K$ is of the form $\oplus_{i=1}^n \mathbb{Z} \cdot \alpha_i$ for some algebraic integers $\alpha_1, \dots, \alpha_n$.

Putting $S = \{\alpha_1, \dots, \alpha_n\}$, we see that $\mathcal{O}_K = R_S$ is an order. It follows easily that $\mathcal{O}_K$ is a *maximal* order in K , in the sense that it is an order which contains all other orders.

We note that $\det(B_S)$ is *independent* of the choice of $\mathbb{Z}$ -basis of $\mathcal{O}_K$ since a change of basis will lead to multiplication by the *square* of the associated change of variable matrix and the change of variable matrix has determinant ± 1 . Thus, it is an “invariant” associated with the field K ; this invariant is called the *discriminant of the field K* and is denoted as D_K .

Let the dual of the order $\mathcal{O}_K$ be $(\mathcal{O}_K)^*$. Note that it contains $\mathcal{O}_K$ and is contained in $(1/D_K)\mathcal{O}_K$. We define the *different* δ_K to be the collection of α in K such that $\alpha \cdot (\mathcal{O}_K)^* \subset \mathcal{O}_K$.

Hermite-Minkowski Theorem

We see easily that $D_{\mathbb{Q}} = 1$.

Hermite proved that given a degree n and a positive integer N , there are at most finitely many number fields of degree n over $\mathbb{Q}$ whose discriminant is at most N .

Minkowski proved an explicit bound which shows that the degree of the field extension is also bounded in terms of the discriminant. This bound implies that the *only* number field K with $D_K = 1$ is $K = \mathbb{Q}$. It also improves Hermite’s theorem to show that given a positive integer N , there are at most finitely many number fields whose discriminant is at most N .